Repaso del teorico

$$n_{medio} = \frac{\text{rapidez de la luz en el vacio}}{\text{rapidez de la luz en el medio}}$$

$$n_m = \frac{c}{v_m}$$

ley de la refracción

$$\frac{n_2}{n_1} = \frac{\operatorname{sen}\theta_1}{\operatorname{sen}\theta_2} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

$$n_1 \operatorname{sen}\theta_1 = n_2 \operatorname{sen}\theta_2$$

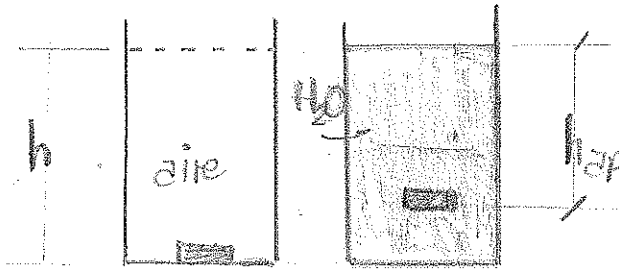
$$n_1 v_1 = n_2 v_2$$

$$n_1 \lambda_1 = n_2 \lambda_2$$

ley de la reflexión

$$\theta_1 = \theta_1'$$

8)



$h = 1\,(m)$

$n_{H_2O} = 1{,}333$

$n_{air} = 1{,}000239$

$n_{air} \approx 1$

$$v_{air} = v_0 = 3 \cdot 10^8\,(m/s) = c$$

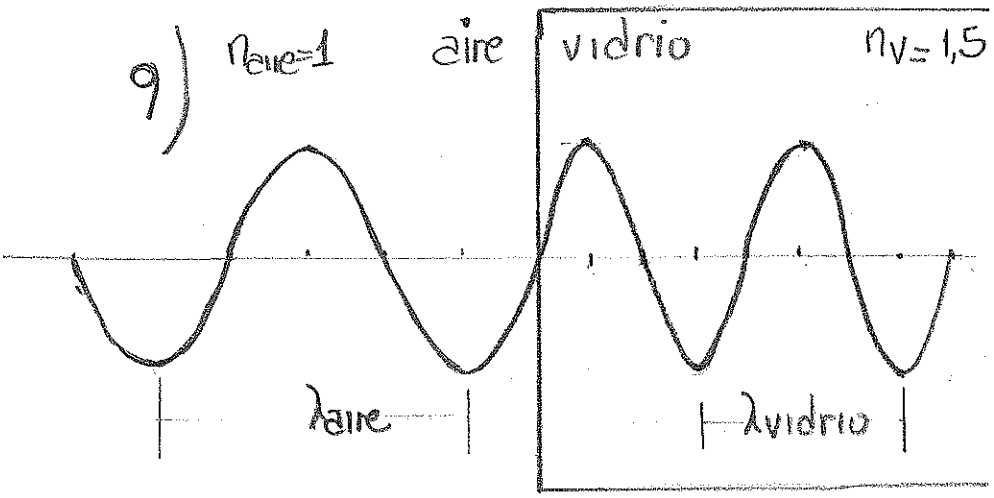
$$v_{air} = \frac{c}{n_{air}} = c = \frac{h}{t_{air}} \qquad t = t_{air} = t_{H_2O}$$

$$v_{H_2O} = \frac{c}{n_{H_2O}} = \frac{h_{apparente}}{t_{H_2O}} = \frac{h_{apparente}}{t}$$

$$t = \frac{h}{c} = \frac{h_{apparente} \cdot n_{H_2O}}{c}$$

$$h_{apparente} = \frac{h}{n_{H_2O}} = \frac{1}{1{,}333} \approx 0{,}75\,(m)$$

9) $n_{aire}=1$ aire | vidrio $n_V=1,5$



$\lambda_{aire} = 632,8\ (nm)$

$v_{aire} = c = 3\ 10^8\ (m/s)$

$v = f.\lambda \begin{cases} v_{aire} = f.\lambda_{aire} \\ v_{vidrio} = f.\lambda_{vidrio} \end{cases}$ la frecuencia f no cambia en la refracción

a) $f = \dfrac{v_{aire}}{\lambda_{aire}} = \dfrac{c}{\lambda_{aire}} = \dfrac{3.10^8\ (m/s)}{632,8\ (nm)} = \dfrac{3.10^8\ \left(\frac{m}{s}\right)}{632,8.10^{-9}\ (m)} =$

$f = 4,74 \cdot 10^{14}\left(\frac{1}{s}\right) = 4,74.10^{14}\ (s^{-1})$

$[f] = (s^{-1}) = (Hertz)$

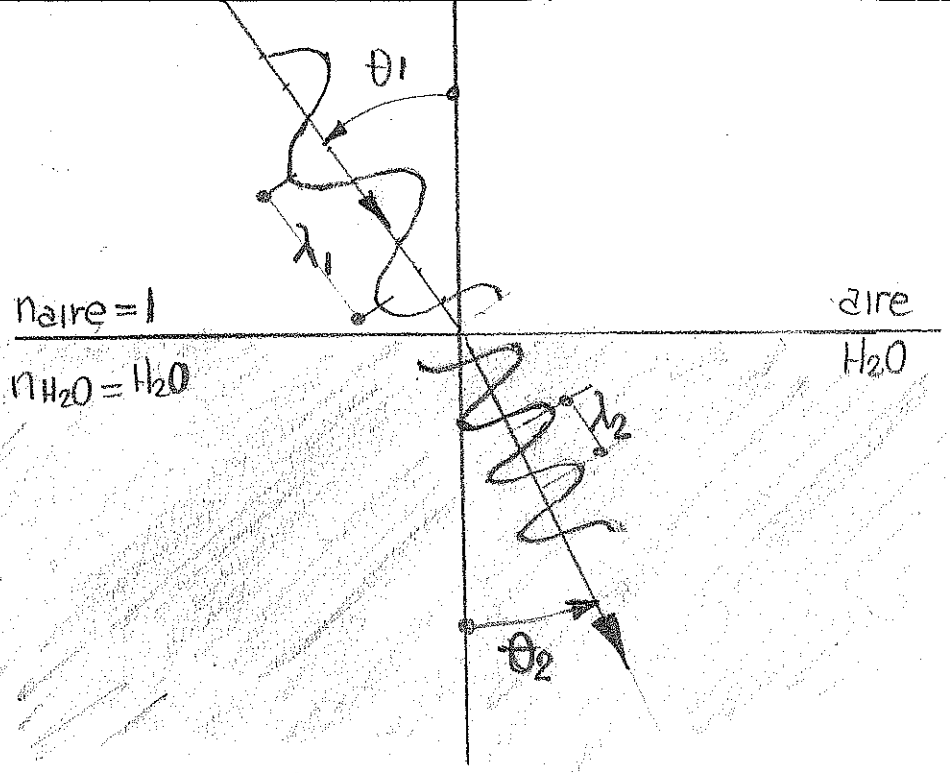
b) $\lambda_{vidrio} \cdot n_{vidrio} = \lambda_{aire} \cdot \cancel{n_{aire}}^{1}$

$\lambda_{vidrio} = \dfrac{\lambda_{aire}}{n_{vidrio}} = \dfrac{632,8\ (nm)}{1,5} = 421,9\ (nm)$

c) $v_{vidrio} \cdot n_{vidrio} = v_{aire} \cdot \cancel{n_{aire}}^{1} = c$

$v_{vidrio} = \dfrac{c}{n_{vidrio}} = \dfrac{3.10^8\ (m/s)}{1,5} = 2.10^8\left(\frac{m}{s}\right)$

10)



$\theta_1 = 35°$

$\theta_2 = ?$

$\lambda_{aire} = 589,5\,(nm)$

$\lambda_{H_2O} = \lambda_2 = ?$

$n_{H_2O} = 1,333$

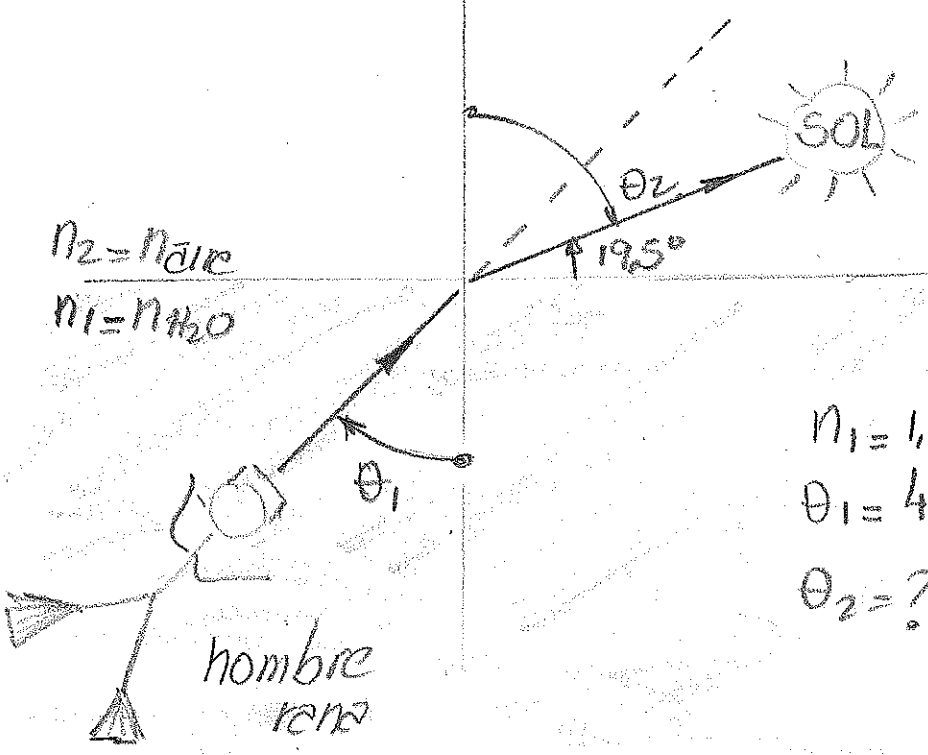
$n_{aire}^{1} \operatorname{sen}\theta_1 = n_{H_2O} \operatorname{sen}\theta_2$

$$\theta_2 = \text{arc sen}\left(\frac{\text{sen}35}{1,333}\right) = 25,5$$

$n_{aire}^{1}\ \lambda_{aire} = n_{H_2O}\ \lambda_{H_2O}$

$$\lambda_{H_2O} = \frac{\lambda_{aire}}{n_{H_2O}} = \frac{589,5\,(nm)}{1,333} = 442,2\,(nm)$$

11)



$n_1 = 1{,}333$

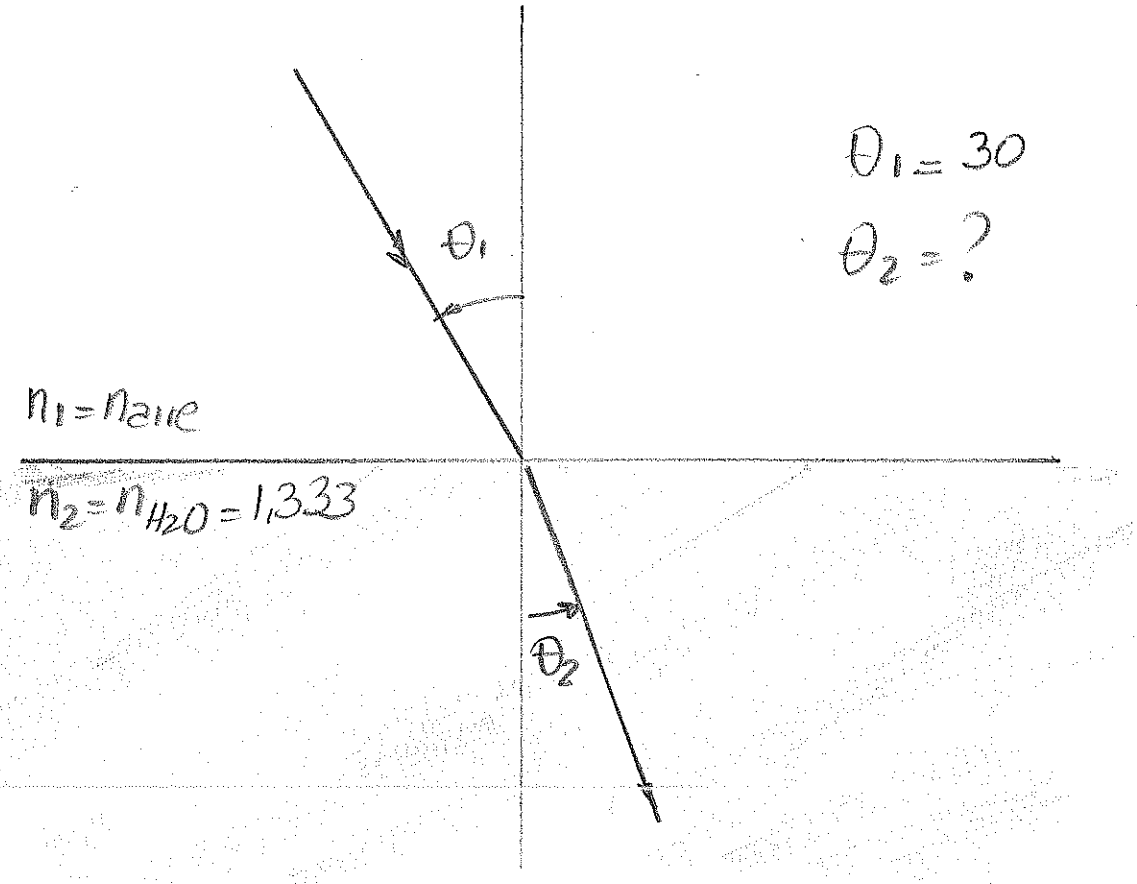
$\theta_1 = 45°$

$\theta_2 = ?$

$$n_1 \operatorname{sen}\theta_1 = n_2 \operatorname{sen}\theta_2$$

$$\theta_2 = \operatorname{arcsen}(1{,}333 \operatorname{sen} 45) = 70{,}5°$$

12)



$\theta_1 = 30$

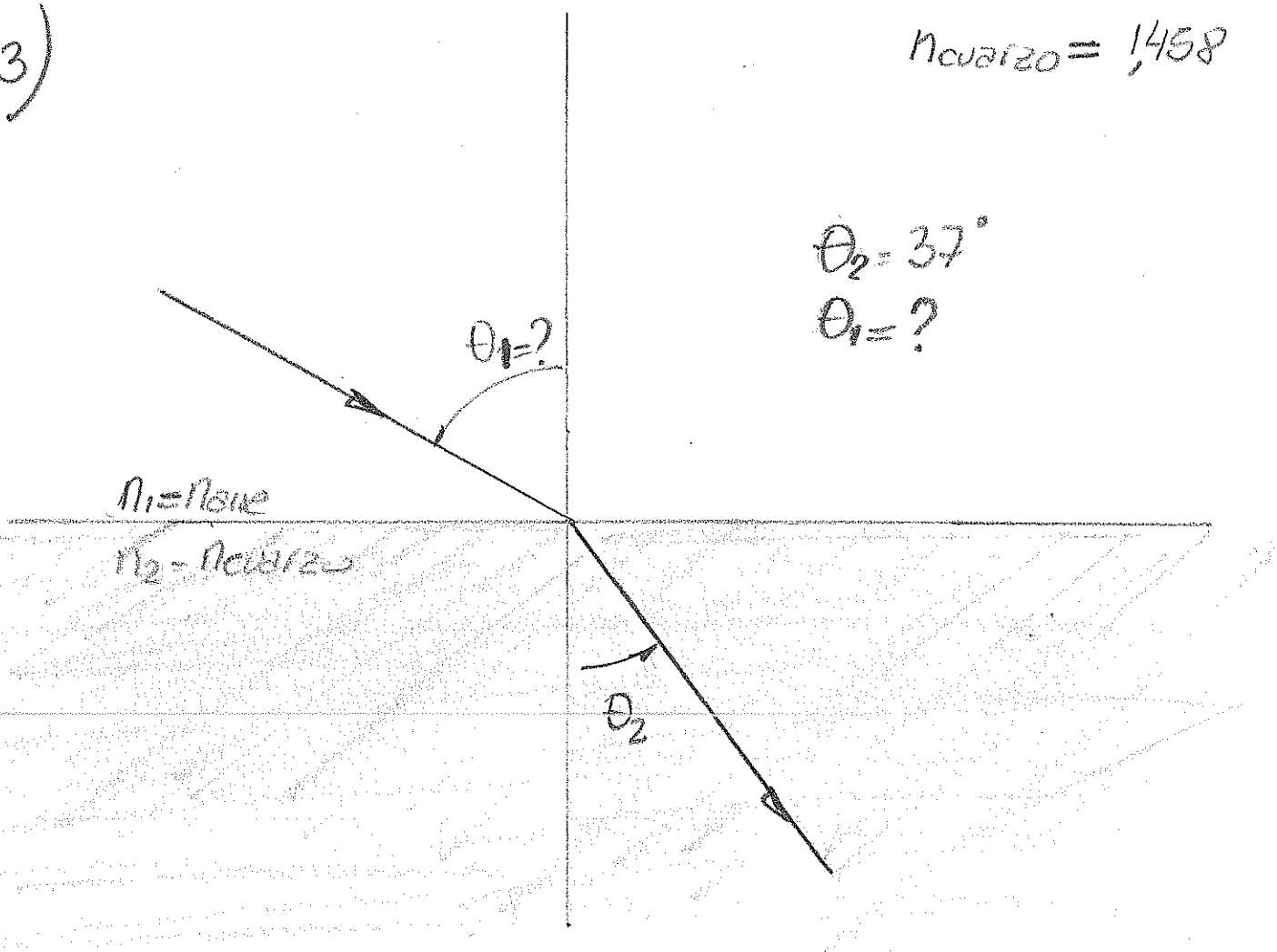
$\theta_2 = ?$

$n_1 \operatorname{sen}\theta_1 = n_2 \operatorname{sen}\theta_2$

$$\theta_2 = \operatorname{arc\,sen}\left(\frac{\operatorname{sen} 30}{1,333}\right) = 22°$$

13)

$n_{cuarzo} = 1{,}458$

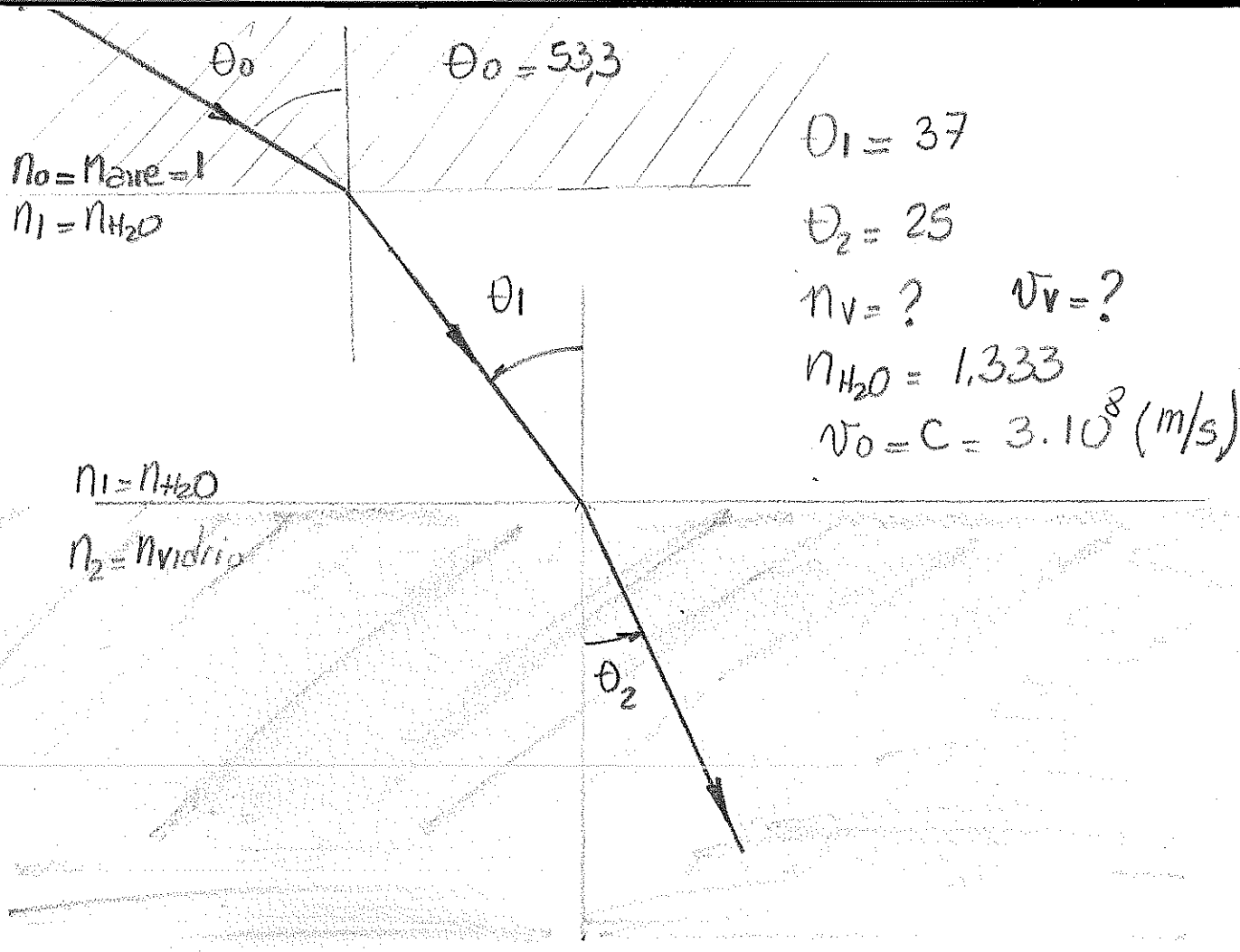


$n_1 \operatorname{sen}\theta_1 = n_2 \operatorname{sen}\theta_2$

$\theta_1 = \operatorname{arc\,sen}(1{,}458 \operatorname{sen} 37)$

$\theta_1 = 61{,}3°$

14)



$\theta_1 = 37$

$\theta_2 = 25$

$n_V = ?$ $\quad v_V = ?$

$n_{H_2O} = 1,333$

$v_0 = c = 3 \cdot 10^8 \ (m/s)$

$$n_1 \cdot \operatorname{sen}\theta_1 = n_2 \cdot \operatorname{sen}\theta_2$$

$$n_2 = n_V = \frac{n_{H_2O} \cdot \operatorname{sen} 37}{\operatorname{sen} 25} = 1,898$$

$$n_0 v_0 = n_1 v_1$$

$$v_1 = v_{H_2O} = \frac{c}{n_{H_2O}} = \frac{3 \cdot 10^8}{1,333} = 2,25 \cdot 10^8 \ (m/s)$$

$$n_1 v_1 = n_2 v_2$$

$$n_{H_2O} v_{H_2O} = n_V \cdot v_V$$

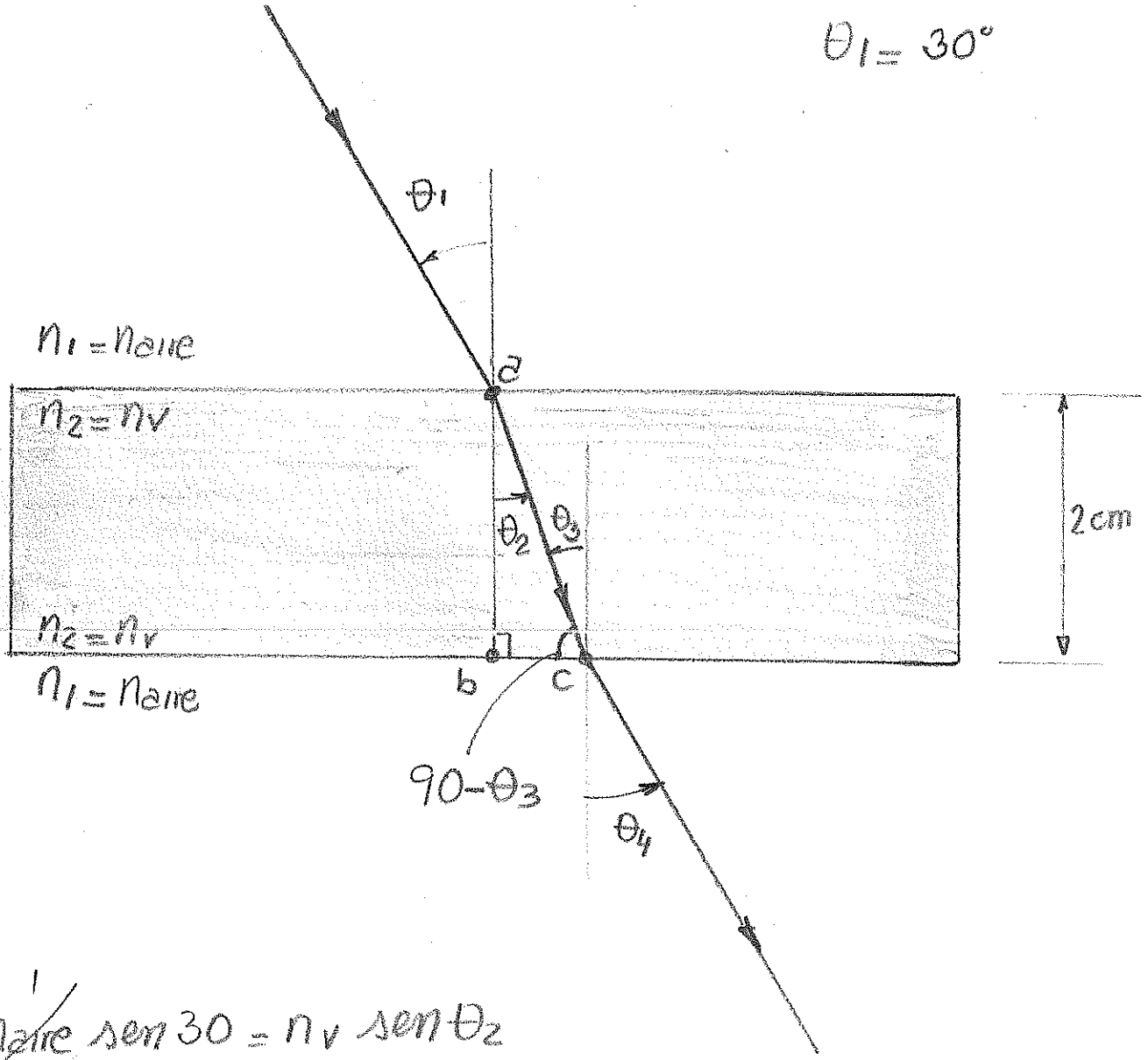
$$n_{H_2O} \frac{c}{n_{H_2O}} = 1,898 \cdot v_V$$

$$v_V = \frac{c}{n_V} = \frac{3 \cdot 10^8}{1,898} = 1,58 \cdot 10^8 \left(\frac{m}{s}\right)$$

15)

$n_V = 1{,}5$

$\theta_1 = 30°$



$n_{ar} \operatorname{sen} 30 = n_V \operatorname{sen} \theta_2$

$\theta_2 = \operatorname{arc\,sen}\left(\frac{\operatorname{sen} 30}{1{,}5}\right) = 19{,}5°$

triângulo $\hat{abc}$: $\theta_2 + 90 + (90 - \theta_3) = 180°$

$\theta_2 = \theta_3 = 19{,}5°$

$n_V \operatorname{sen} 19{,}5° = n_{ar} \operatorname{sen} \theta_4$

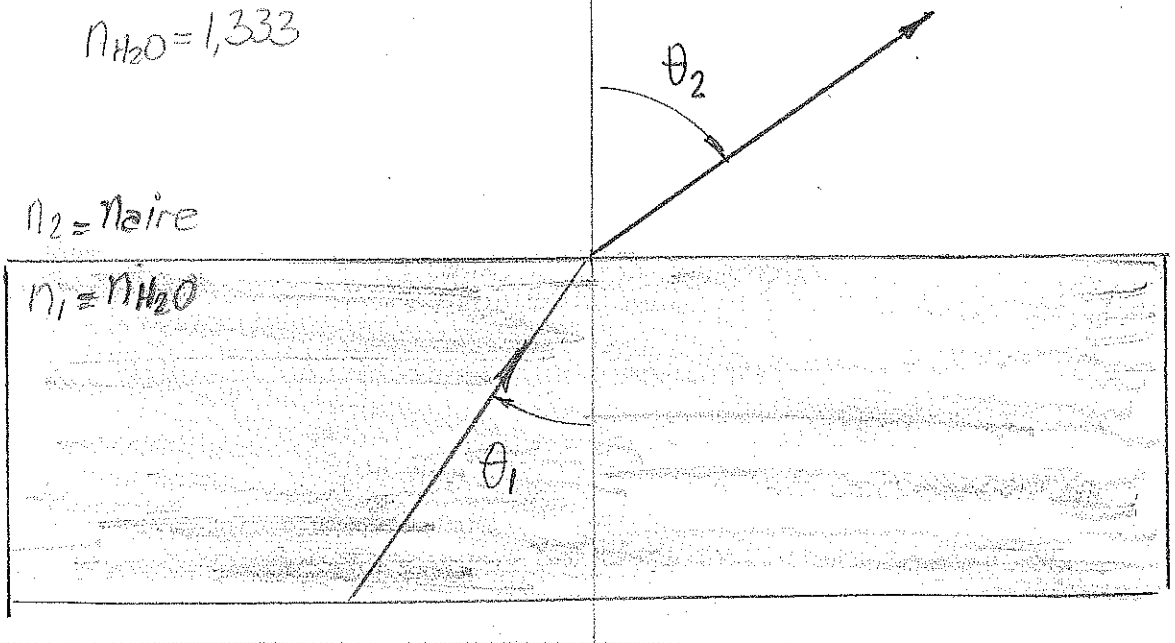
$\theta_4 = \operatorname{arc\,sen}(1{,}5 \cdot \operatorname{sen} 19{,}5) = 30°$

$\theta_1 = \theta_4 = 30°$

16) $\theta_1 = 37°$

$\theta_2 = ?$

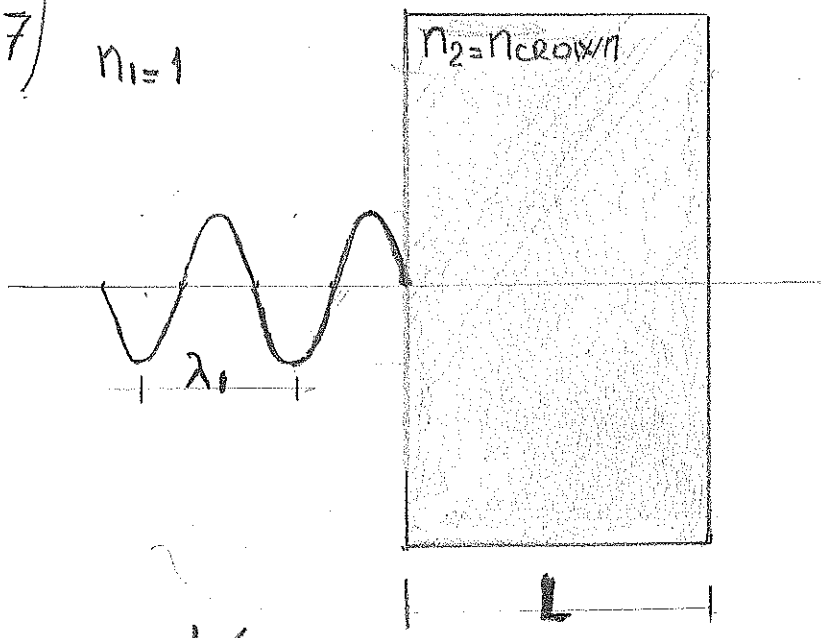
$n_{H_2O} = 1{,}333$



$n_{H_2O} \cdot \text{sen } 37° = n_{ar} \text{ sen } \theta_2$

$\theta_2 = \text{arc sen} \left(1{,}333 \cdot \text{sen } 37° \right) = 53{,}3°$

17)



$n_2 = n_{CROWN} = 1{,}52$

$n_0 = n_1 = 1$

$\lambda_1 = \lambda_0 = 589\,(nm)$

$v_{CROWN} = ?$

$c = v_0 = v_1 = 3\,10^8\,(m/s)$

$\lambda_2 = \lambda_{CROWN} = ?$

$L = 100\,\lambda_{CROWN}$

$n_1 v_1 = n_2 v_2$

$c = 1{,}52\, v_2$

$$v_2 = \frac{c}{1{,}52} = \frac{3 \cdot 10^8\,(m/s)}{1{,}52} = 1{,}97 \cdot 10^8\,(m/s)$$

$n_1 \cdot \lambda_1 = n_2 \cdot \lambda_2$

$\lambda_0 = 1{,}52 \cdot \lambda_{CROWN}$

$$\lambda_{CROWN} = \frac{\lambda_0}{1{,}52} = \frac{589\,(nm)}{1{,}52} = 387{,}5\,(nm)$$

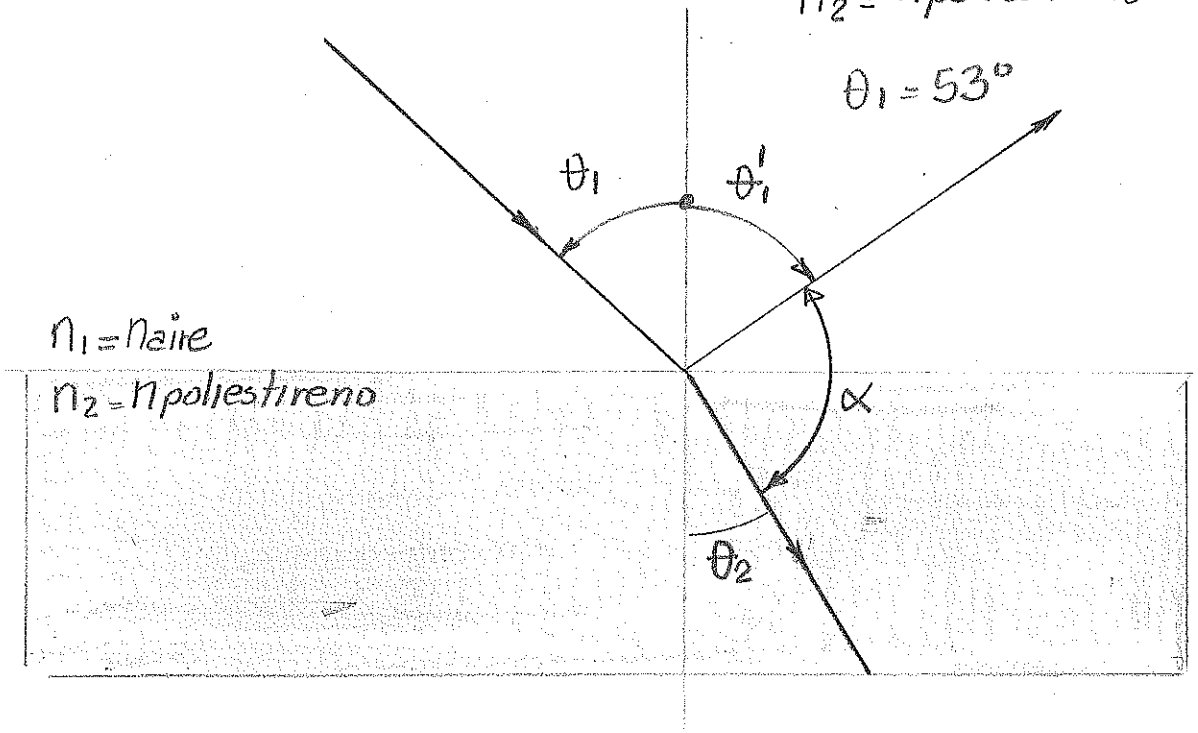
$L = 100 \cdot \lambda_{CROWN} = 10^2 \cdot 387{,}5 \cdot 10^{-9}\,(m)$

$= 387{,}5 \cdot 10^{-7}\,(m)$

$= 38{,}75\,10^{-6}\,(m) = 38{,}75\,(\mu m)$

18)

$n_2 = n_{poliestireno} = 1,49$



$$n_1 \operatorname{sen}\theta_1 = n_2 \operatorname{sen}\theta_2$$

$$\operatorname{sen}53 = 1,49 \operatorname{sen}\theta_2$$

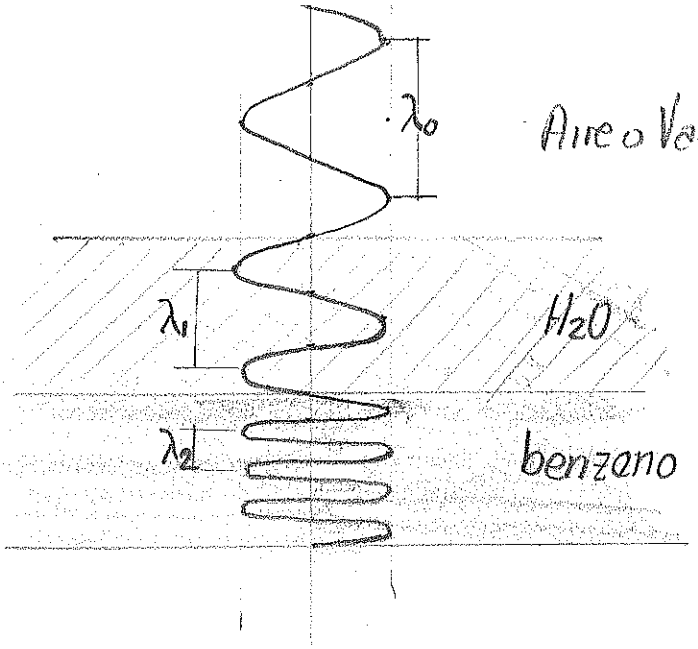
$$\theta_2 = \operatorname{arcsen}\left(\frac{\operatorname{sen}53}{1,49}\right) = 32,4°$$

$$\theta_1 = \theta_1' = 53°$$

$$\theta_1' + \alpha + \theta_2 = 180°$$

$$\alpha = 180 - \theta_1' - \theta_2 = 180 - 53° - 32,4 = 94,6$$

19)



$\lambda_1 = \lambda_{H_2O} = 438\ (nm)$

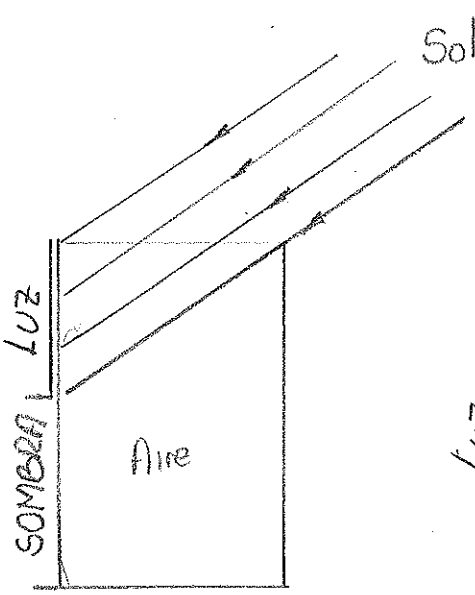
$\lambda_2 = \lambda_{benzeno} = 390\ (nm)$

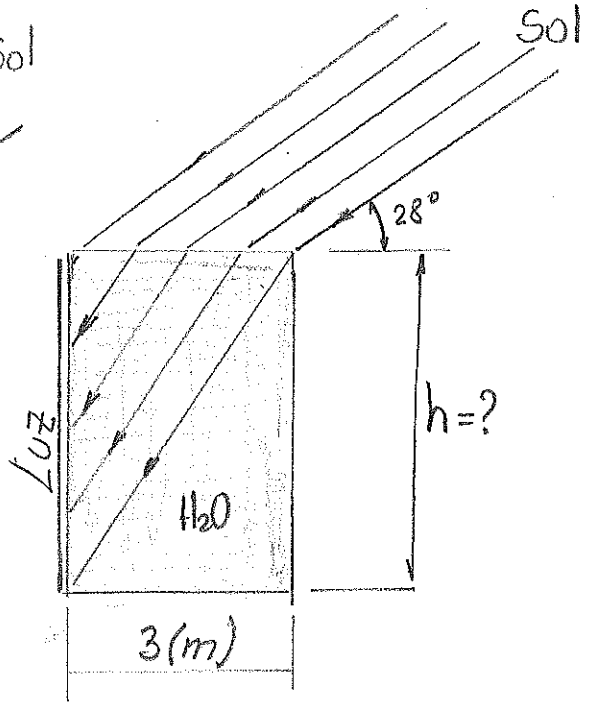
$$n_{H_2O} \cdot \lambda_{H_2O} = n_{benzeno} \cdot \lambda_{benzeno}$$

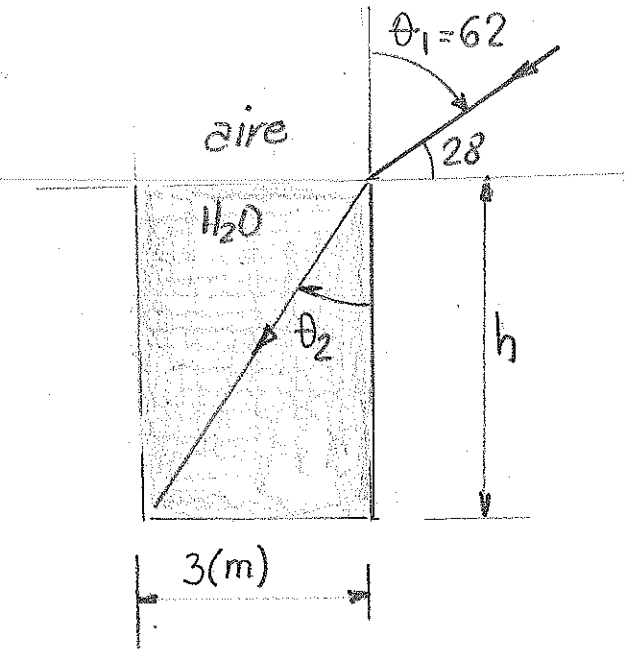
$$\frac{n_{H_2O}}{n_{benzeno}} = \frac{\lambda_{benzeno}}{\lambda_{H_2O}} = \frac{390\ (nm)}{438\ (nm)} = 0{,}89$$

índice de refracción relativo entre el H_2O y el benzeno $= \dfrac{n_{H_2O}}{n_{benzeno}} = 0{,}89$

20)







$$n_{aire} \operatorname{sen}\theta_1 = n_{H_2O} \operatorname{sen}\theta_2$$

$$\theta_2 = \operatorname{arc\,sen}\left(\frac{\operatorname{sen} 62}{1{,}333}\right) = 41{,}48°$$

$$\operatorname{tg}\theta_2 = \frac{3(m)}{h}$$

$$h = \frac{3(m)}{\operatorname{tg} 41{,}48} = 3{,}39(m)$$